

Data Science for Biologists: A Hands-On Introduction with R and Bioconductor

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6 August 2026

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Abstract

Background. Biomedical researchers are increasingly expected to analyze their own data reproducibly, yet most have had no formal training in statistics or data science, and generic “intro to R” materials teach syntax without the conceptual grounding that biological data analysis requires.

Methods. *The RBioc Book* is an open-access, fully executable textbook, built as a Quarto book on the `knitr` engine so that every chapter runs its own R code at render time and prose, code, and output are always in sync. It is designed around adult-learning principles, carries biological framing from the first chapters, and builds a cumulative conceptual scaffold including R programming, data structures, and statistical reasoning that the genomics and single-cell chapters later reuse. It is openly licensed (CC-BY 4.0), versioned, and archived with a citable DOI.

Results. The book takes a reader from a first line of R to the analysis of real genomic data with Bioconductor, teaching against public datasets. It has been taught in four formal course offerings (~100 trainees, 2025–2026) and, as an online resource, recorded 9,098 page views across 62 pages from 1,475 unique readers between May 2024 and August 2026, with per-chapter active-engagement times indicating readers work through rather than skim the material. Instructors can adopt a chapter as a lab, fork it for a local course, or assign it for self-paced study; every result is reproducible by re-running the code.

Conclusions. *The RBioc Book* is a complete, classroom-tested, openly-licensed teaching resource that closes the concept-and-reproducibility gap for biologists endeavoring to understand data science, statistics, and reproducible computational

research. It is also a template for large-scale executable, community-maintained teaching materials in the Bioconductor ecosystem.

Introduction

Biologists increasingly need to understand statistical and data science principles, but many biologists have limited formal coursework in statistics or data science (Drnevich et al. 2025). At the same time they are expected to drive analyses of their own data — and to do so reproducibly. Many generic “intro to R” courses show *how* to perform an analysis but do not motivate it with concept and rationale, leaving learners able to copy a workflow but not to reason about it or adapt it to their own problems.

The RBioc Book — the resource described here, published under that name at <https://seandavi.github.io/RBiocBook> and archived under it at Zenodo — is an open-access, executable textbook that takes a reader from their first line of R (R Core Team 2024) to the analysis of real genomic data with Bioconductor (Huber et al. 2015). It is written for biomedical researchers who know the biology and have been introduced to programming, statistics, and computation but want a more formal and thorough treatment of the concepts and skills; a biology primer makes it usable from the other direction as well, for computational readers approaching biological data for the first time. Rather than teaching programming in the abstract and leaving the biology for “later,” the book carries biological framing from the first chapters and builds the conceptual scaffolding that the genomics chapters later depend on. Because the book is fully executable and openly licensed, an instructor can adopt a chapter as a lab, fork it for a local course, or assign it for self-paced study, and a learner can reproduce every result by running the same code.

Relation to existing resources. Excellent open teaching resources already exist at both ends of this path, and the gap *The RBioc Book* addresses is the span between them. *R for Data Science* (Wickham, Çetinkaya-Rundel, and Grolemund 2023) is the standard introduction to data analysis in R, but it teaches against general-purpose datasets and does not carry the reader toward biological data or the Bioconductor classes that biology requires. *Modern Statistics for Modern Biology* (Holmes and Huber 2019) treats statistical reasoning for biological data with far greater depth than we attempt, but assumes a reader already fluent in R and comfortable with mathematical statistics. *Orchestrating Single-Cell Analysis with Bioconductor* (Amezquita et al. 2020) is the reference work for one domain and, by design, begins where a competent Bioconductor user begins. A learner starting from no R at all must therefore assemble a path across three resources with different notational conventions, different assumed backgrounds, and no shared running example — and, in our teaching experience, most stall at the seams rather than inside any one text.

The RBioc Book is deliberately not a deeper treatment of any of these. Its contri-

bution is continuity: one notation, one cumulative scaffold, and one executable artifact spanning from a first line of R to real Bioconductor genomics, with the biological motivation present from the beginning rather than deferred. It is intended to precede rather than replace the resources above, and it points to them where a reader should go deeper. This positioning also reflects a broader need identified across the Bioconductor teaching community (Drnevich et al. 2025).

Methods

Implementation

The book is built as a Quarto book (Allaire et al. 2024) so that rendering *executes* the R code in each chapter: the prose, the code, and the output the reader sees are produced from the same source and are always in sync. This executable design is at the core of the book. It makes the entire book a reproducible artifact rather than a static document, and it means that keeping the book correct is the same act as keeping the code runnable.

The material is organized around adult-learning principles (Knowles, Holton III, and Swanson 2015). Each chapter opens with explicit learning objectives, motivates concepts before formalizing them, and pairs worked examples with exercises and solutions. A consistent callout taxonomy (objectives, notes, warnings, exercises) gives the reader predictable signposts throughout. The narrative is intentionally cumulative: early chapters earn the vocabulary and data structures that the statistics, machine-learning, and genomics chapters reuse. The genomics material is taught against real, public datasets and the Bioconductor classes practitioners actually use — including the genomic-interval infrastructure of **GenomicRanges** (Lawrence et al. 2013) — so the skills transfer directly to the reader’s own analyses. Data manipulation and visualization are taught with the **tidyverse** (Wickham et al. 2019) and **ggplot2** (Wickham 2016) alongside base R, so that readers encounter the idioms actually used in practice rather than a single house style.

Operation

Minimal requirements. Reading the book online requires only a web browser (<https://seandavi.github.io/RBiocBook>). Building or extending it locally requires R with the packages declared in the repository’s **DESCRIPTION**, and Quarto; the **knitr** engine executes each chapter’s R at render time.

Workflow. From a clone of the repository:

```
quarto render                                # all formats (html, pdf, epub) into _book/
quarto preview                              # live-reload server while editing
quarto render <file>.qmd --to html          # single chapter, fast iteration
```

Adoption modes. Because each chapter is a self-contained, executable `.qmd` document, an instructor can (i) assign the online edition for self-paced study, (ii) adopt an individual chapter as a lab, or (iii) fork the repository and adapt chapters for a local course, re-rendering to regenerate every result. The book is versioned and each release is archived with a citable DOI, so a course can pin to a specific version.

Use cases

The RBioc Book is designed to support both synchronous and asynchronous teaching: learners ingest material at their own pace, freeing instructors to engage participants directly rather than lecture syntax.

Executable chapter (input $\rightarrow$ output). The single-cell RNA-seq chapter is representative. It loads a published mouse-brain dataset as a `SingleCellExperiment`, log-normalizes the counts, selects highly variable genes, runs PCA and UMAP, clusters the cells, and identifies and visualizes the marker genes that define each cluster — producing every table and figure inline at render time, with no hidden state, so a reader reproduces the chapter by rendering it. It also shows the cumulative design concretely: `SingleCellExperiment` is introduced as an extension of the `SummarizedExperiment` class taught in an earlier chapter, and the chapter tells the reader to revisit that chapter first if the container’s structure is not fresh. This input (chapter source + declared R environment) $\rightarrow$ output (rendered prose, code, and result) round-trip is the unit the book is built from.

Formal course use. The book has been taught in four formal offerings — the Cold Spring Harbor Laboratory data-analysis course (2025 and 2026) and the BigCARE program (2025 and 2026), each with roughly 25 participants per session (about 100 trainees over two years).

Open readership. Beyond the classroom, between May 2024 and August 2026 the online edition recorded 9,098 page views across 62 distinct pages from 1,475 unique readers. The reader count is Google Analytics 4’s own de-duplicated figure for pages under the book’s path prefix, so a reader who works through several chapters is counted once; it is not a sum of per-page counts, which would over-count the same reader to 4,933. For scale, the landing page alone accounts for 990 of those readers. Engagement is substantial where the material is hardest: the single-cell chapter averaged 7.4 minutes of active engagement per reader, the `ggplot2` and simulation chapters 6.7 and 6.0 minutes, and the R-basics chapter 4.4 minutes, indicating that readers work through the material rather than skim it. The underlying export is archived in the repository (`paper/data/`), and the script that produced it is included (`paper/ga4_pull.py`), so every figure here can be recomputed. The book continues to be developed in the open and contains social coding best practice guidance to accept contributions.

Summary

The RBioc Book closes the gap between “how to run R” and “how to reason about, and reproducibly perform, biological data analysis,” in one continuous path from a first line of R to real Bioconductor genomics. Its executable, openly-licensed, version-archived design makes it a reusable teaching resource — adoptable whole or in part — and a template for community-maintained, reproducible teaching materials in the Bioconductor ecosystem.

Scope and limitations. The book is a teaching resource, not a methods reference: it teaches the reasoning and the standard workflow for each topic rather than surveying alternatives, and it deliberately stops short of specialized or fast-moving areas — among them multi-omic integration, spatial transcriptomics, and workflow-level pipeline orchestration — that would date quickly or require prerequisites the book does not build. The evidence of use reported here is adoption and engagement data; the book has not been evaluated against learning outcomes in a controlled setting, and such a study would be the natural next step for assessing pedagogical effect rather than reach. Being executable also carries an ongoing maintenance obligation: chapters are re-rendered against current package versions, and upstream Bioconductor changes require corresponding updates — a cost the design converts from silent documentation rot into a visible build failure, but does not eliminate.

Data availability

The book teaches against publicly available datasets, cited and linked in place within the relevant chapters. Website usage figures reported here were derived from Google Analytics 4, restricted to pages under the book’s path prefix, over May 2024–August 2026. Both the export used for the reported totals and the script that generated it are included in the source repository (`paper/data/` and `paper/ga4_pull.py`), so the figures can be reproduced or recomputed for a later window.

Software and resource availability

- **Source code / book repository:** <https://github.com/seandavi/RBiocBook>
- **Live book:** <https://seandavi.github.io/RBiocBook>
- **Archived version (concept DOI):** <https://doi.org/10.5281/zenodo.20574829>
- **License:** Original content (prose, original figures, example code) is released under CC-BY 4.0; example code is additionally reusable as such. Third-party figures, datasets, and adapted passages retain their own open licenses, each documented on the book’s *Contributors & acknowledgments* page.
- **Machine-readable citation:** `CITATION.cff` in the repository.

Author contributions

Lori Shepherd Kern — authored *Base R versus the tidyverse*, *Control statements*, *Factors*, and *Organizing, saving, and loading your work*, with edits throughout. Sean Davis — conceived the book; primary author and editor; wrote most chapters. Both authors reviewed and approved the manuscript.

Competing interests

No competing interests were disclosed.

Grant information

This work was supported in part by the US National Institutes of Health / National Cancer Institute Informatics Technology for Cancer Research (ITCR) program through the Bioconductor U24 award (U24 CA289073).

Acknowledgements

We thank Martin Morgan (Bioconductor Core Team) for contributions to the *Base R versus the tidyverse* chapter and the *BRFSS case study*; the Cold Spring Harbor Laboratory and BigCARE course participants whose questions shaped the material; the Bioconductor community; and the contributors credited on the book's *Contributors & acknowledgments* page.

References

- Allaire, J. J., Charles Teague, Carlos Scheidegger, Yihui Xie, and Christophe Dervieux. 2024. “Quarto.” <https://doi.org/10.5281/zenodo.5960048>.
- Amezquita, Robert A, Aaron T L Lun, Etienne Becht, Vince J Carey, Lindsay N Carpp, Ludwig Geistlinger, Federico Marini, et al. 2020. “Orchestrating Single-Cell Analysis with Bioconductor.” *Nature Methods* 17 (2): 137–45. <https://doi.org/10.1038/s41592-019-0654-x>.
- Drnevich, Jenny, Frederick J Tan, Fabricio Almeida-Silva, Robert Castelo, Aedin C Culhane, Sean Davis, Maria A Doyle, et al. 2025. “Learning and Teaching Biological Data Science in the Bioconductor Community.” *PLoS Computational Biology*. <https://arxiv.org/abs/2410.01351>.
- Holmes, Susan, and Wolfgang Huber. 2019. *Modern Statistics for Modern Biology*. Cambridge, UK: Cambridge University Press. <https://www.huber.embl.de/msmb/>.
- Huber, Wolfgang, Vincent J Carey, Robert Gentleman, Simon Anders, Marc Carlson, Benilton S Carvalho, Hector Corrada Bravo, et al. 2015. “Orchestrating High-Throughput Genomic Analysis with Bioconductor.” *Nature Methods* 12 (2): 115–21. <https://doi.org/10.1038/nmeth.3252>.

- Knowles, Malcolm S, Elwood F Holton III, and Richard A Swanson. 2015. *The Adult Learner: The Definitive Classic in Adult Education and Human Resource Development*. 8th ed. Routledge.
- Lawrence, Michael, Wolfgang Huber, Hervé Pagès, Patrick Aboyoun, Marc Carlson, Robert Gentleman, Martin T Morgan, and Vincent J Carey. 2013. “Software for Computing and Annotating Genomic Ranges.” *PLoS Computational Biology* 9 (8): e1003118. <https://doi.org/10.1371/journal.pcbi.1003118>.
- R Core Team. 2024. “R: A Language and Environment for Statistical Computing.” Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York. <https://ggplot2-book.org/>.
- Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D’Agostino McGowan, Romain François, Garrett Grolemund, et al. 2019. “Welcome to the Tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.
- Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. 2023. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. 2nd ed. O’Reilly Media. <https://r4ds.hadley.nz/>.